

```
henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % clear
henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % gcc -o countnames countnames.c -Wall -Werror
henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % gcc -o shell shell.c -Wall -Werror
henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % ./shell
% ./countnames test/names.txt test/namesB.txt test/names1.txt test/names2.txt test/names_long.txt test/names_long_redundant.txt test/names_long_redundant1.txt test/names_long_redundant2.txt test/names_long_redundant3.txt test/empty.txt test/emptylines.txt test/tiny.txt
Child 87490 terminated normally with exit code: 0
Child 87488 terminated normally with exit code: 0
Child 87487 terminated normally with exit code: 0
Child 87486 terminated normally with exit code: 0
Child 87483 terminated normally with exit code: 0
Child 87482 terminated normally with exit code: 0
Child 87481 terminated normally with exit code: 0
Child 87479 terminated normally with exit code: 0
Child 87484 terminated normally with exit code: 0
Child 87489 terminated normally with exit code: 0
Child 87485 terminated normally with exit code: 0
Child 87480 terminated normally with exit code: 0
% ./countnames test/names.txt test/names.txt
Child 87517 terminated normally with exit code: 0
Child 87518 terminated normally with exit code: 0
% ./countnames test/names1.txt test/nope.txt test/names2.txt
Child 87537 terminated normally with exit code: 0
Child 87538 terminated normally with exit code: 1
Child 87539 terminated normally with exit code: 0
%
```

```
henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % clear
rm -f [0-9]*.out [0-9]*.err
./shell
% ./countnames test/names.txt test/namesB.txt test/names1.txt test/names2.txt test/names_long.txt test/names_long_redundant.txt test/names_long_redundant1.txt test/names_long_redundant2.txt test/names_long_redundant3.txt test/empty.txt test/emptylines.txt test/tiny.txt
Child 88520 terminated normally with exit code: 0
Child 88519 terminated normally with exit code: 0
Child 88515 terminated normally with exit code: 0
Child 88514 terminated normally with exit code: 0
Child 88512 terminated normally with exit code: 0
Child 88521 terminated normally with exit code: 0
Child 88522 terminated normally with exit code: 0
Child 88517 terminated normally with exit code: 0
Child 88516 terminated normally with exit code: 0
Child 88513 terminated normally with exit code: 0
Child 88518 terminated normally with exit code: 0
Child 88511 terminated normally with exit code: 0
% █
```

```
henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % clear
for f in [0-9]*.out; do echo "=== $f ($(wc -l < $f) names) ==="; head -4 $f; done
echo "---- warnings ----"
cat [0-9]*.err
=== 88511.out (      4 names) ===
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
=== 88512.out (      4 names) ===
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
=== 88513.out (      1 names) ===
Tom Wu: 3
=== 88514.out (      2 names) ===
Jenn Xu: 2
Tom Wu: 1
=== 88515.out (    101 names) ===
MARY SMITH: 1
: 99
PATRICIA JOHNSON: 1
LINDA WILLIAMS: 1
=== 88516.out (     91 names) ===
John Smith: 1
Emily Davis: 1
Michael Johnson: 1
Sophia Lee: 1
=== 88517.out (     40 names) ===
MARY SMITH: 1
PATRICIA JOHNSON: 1
LINDA WILLIAMS: 1
BARBARA JONES: 1
=== 88518.out (     44 names) ===
STEPHANIE MITCHELL: 1
CAROLYN PEREZ: 1
CHRISTINE ROBERTS: 1
MARIE TURNER: 1
=== 88519.out (     41 names) ===
ANNE J POWELL: 1
JACQUELINE K LONG: 1
WANDA M PATTERSON: 1
BONNIE HUGHES: 1
=== 88520.out (      0 names) ===
=== 88521.out (      2 names) ===
Ann Lee: 2
Bo Chen: 1
=== 88522.out (      1 names) ===
```



```
Tom Wu: 1
=== 88515.out (      101 names) ===
MARY SMITH: 1
      : 99
PATRICIA JOHNSON: 1
LINDA WILLIAMS: 1
=== 88516.out (       91 names) ===
John Smith: 1
Emily Davis: 1
Michael Johnson: 1
Sophia Lee: 1
=== 88517.out (       40 names) ===
MARY SMITH: 1
PATRICIA JOHNSON: 1
LINDA WILLIAMS: 1
BARBARA JONES: 1
=== 88518.out (       44 names) ===
STEPHANIE MITCHELL: 1
CAROLYN PEREZ: 1
CHRISTINE ROBERTS: 1
MARIE TURNER: 1
=== 88519.out (       41 names) ===
ANNE J POWELL: 1
JACQUELINE K LONG: 1
WANDA M PATTERSON: 1
BONNIE HUGHES: 1
=== 88520.out (         0 names) ===
=== 88521.out (         2 names) ===
Ann Lee: 2
Bo Chen: 1
=== 88522.out (         1 names) ===
Tom Wu: 2
--- warnings ---
```

```
Warning - file test/names.txt line 2 is empty.
Warning - file test/names.txt line 5 is empty.
Warning - file test/namesB.txt line 2 is empty.
Warning - file test/namesB.txt line 5 is empty.
Warning - file test/names1.txt line 3 is empty.
Warning - file test/names_long_redundant.txt line 97 is empty.
Warning - file test/names_long_redundant1.txt line 2 is empty.
Warning - file test/names_long_redundant1.txt line 4 is empty.
Warning - file test/names_long_redundant1.txt line 6 is empty.
Warning - file test/names_long_redundant1.txt line 8 is empty.
Warning - file test/emptylines.txt line 2 is empty.
Warning - file test/emptylines.txt line 5 is empty.
```

henrylee@Henrys-MacBook-Pro Assignment 2 FA26 %

```
henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % clear
```

```
rm -f [0-9]*.out [0-9]*.err
```

```
./shell
```

```
% ./countnames test/names.txt test/names.txt
```

```
Child 89009 terminated normally with exit code: 0
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Child 89008 terminated normally with exit code: 0
```

```
% █
```

```
henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % clear
rm -f [0-9]*.out [0-9]*.err
./shell
% ./countnames test/names.txt test/names.txt
Child 89009 terminated normally with exit code: 0
Child 89008 terminated normally with exit code: 0
% %
```

henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % ls [0-9]\*.out

```
cat [0-9]*.out
89008.out      89009.out
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
```

```
henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % clear
rm -f [0-9]*.out [0-9]*.err
./shell
% ./countnames test/names1.txt test/nope.txt test/names2.txt
Child 89256 terminated normally with exit code: 1
Child 89257 terminated normally with exit code: 0
Child 89255 terminated normally with exit code: 0
% %
henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % cat [0-9]*.err
wc -c [0-9]*.out
Warning - file test/names1.txt line 3 is empty.
error: cannot open file test/nope.txt
    10 89255.out
     0 89256.out
    21 89257.out
    31 total
henrylee@Henrys-MacBook-Pro Assignment 2 FA26 %
```

```
● henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % clear
rm -f [0-9]*.out [0-9]*.err
echo "--- SERIAL: four large files, one after another ---"
time ( for f in test/big1.txt test/big2.txt test/big3.txt test/big4.txt; do ./countnames $f; done )

echo "--- PARALLEL: the same four through the shell ---"
time ( echo "./countnames test/big1.txt test/big2.txt test/big3.txt test/big4.txt" | ./shell )
--- SERIAL: four large files, one after another ---
( for f in test/big1.txt test/big2.txt test/big3.txt test/big4.txt; do; $f; ) 0.37s user 0.63s system 22% cpu 4.452 total
--- PARALLEL: the same four through the shell ---
% Child 89819 terminated normally with exit code: 0
Child 89822 terminated normally with exit code: 0
Child 89821 terminated normally with exit code: 0
Child 89820 terminated normally with exit code: 0
% ( echo "./countnames test/big1.txt test/big2.txt test/big3.txt test/big4.txt" ) 1.58s user 12.53s system 378% cpu 3.725 total
○ henrylee@Henrys-MacBook-Pro Assignment 2 FA26 % rm -f [0-9]*.out [0-9]*.err
./shell
% ./countnames test/big1.txt test/tiny.txt test/names1.txt test/names2.txt
Child 89950 terminated normally with exit code: 0
Child 89952 terminated normally with exit code: 0
Child 89951 terminated normally with exit code: 0
Child 89949 terminated normally with exit code: 0
```